GRADE 11 TECHPRO ACTIVITY SHEET

RESPONSIVE IMAGE GALLERY USING FLEXBOX

ICT / COMPUTER PROGRAMMING | GRADE 11 | VISUAL STUDIO CODE

Name: Ramo, Jeric Y. Section: 11 - Bell Date: 11 / 12 / 26

LEARNING OBJECTIVES

Create an image gallery using semantic HTML elements.

Use display: flex, flex-wrap, justify-content, and gap to arrange gallery cards.

Style images using width, height, and object-fit.

Use a media query to adapt the gallery to smaller screens.

Provide meaningful alternative text (alt text) for images.

WHAT YOU NEED

Computer with Visual Studio Code and a web browser.

Create a folder named Responsive_Image_Gallery and a file named index.html.

Copy the starter code below, save it, and open index.html in a browser.

The sample image links are placeholders and need an internet connection. Replace them with your own image files if preferred.

CSS CONCEPTS

display: flex turns an element into a flex container; its direct children become flex items.

flex-wrap: wrap allows cards to move onto another row when space is limited.

justify-content: center centers cards along the main axis. gap creates consistent spacing.

flex: 1 1 220px lets a card grow, shrink, and start with a 220-pixel basis.

max-width prevents a card from becoming too wide.

width: 100% makes an image fill its card width. object-fit: cover fills the image area while preserving proportions; edges may be cropped.

@media (max-width: 600px) applies rules when the viewport is 600 pixels wide or narrower.

STARTER CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>TechPro Image Gallery</title>
<style>
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #f5f5f5; }
header { background: #7b1530; color: white; text-align: center; padding: 24px; }
.gallery { display: flex; flex-wrap: wrap; justify-content: center; gap: 16px; max-width: 1100px; margin: auto; padding: 20px; }
.gallery-item { flex: 1 1 220px; max-width: 320px; background: white; border: 2px solid #7b1530; border-radius: 8px; overflow: hidden; }
.gallery-item img { width: 100%; height: 200px; object-fit: cover; display: block; }
.caption { padding: 12px; text-align: center; }
footer { background: #222; color: white; text-align: center; padding: 14px; }
@media (max-width: 600px) {
  .gallery { flex-direction: column; align-items: center; }
  .gallery-item { width: 100%; max-width: 420px; }
}
</style>
</head>
<body>
<header><h1>TechPro Image Gallery</h1><p>Responsive Image Gallery using Flexbox</p></header>
<main>
<section class="gallery">
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Computer+Lab" alt="Computer laboratory">
<div class="caption">Computer Laboratory</div>
</article>
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Programming" alt="Programming activity">
<div class="caption">Programming Activity</div>
</article>
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Robotics" alt="Robotics project">
<div class="caption">Robotics Project</div>
</article>
```

```
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Networking" alt="Networking lesson">
<div class="caption">Networking Lesson</div>
</article>
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Web+Design" alt="Web design activity">
<div class="caption">Web Design</div>
</article>
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=ICT+Projects" alt="ICT student projects">
<div class="caption">ICT Student Projects</div>
</article>
</section>
</main>
<footer>Grade 11 TechPro | Responsive Image Gallery</footer>
</body>
</html>
```

ACTIVITY 1 — BUILD AND VIEW

1. Open the folder in Visual Studio Code and open index.html.
2. Study the HTML structure: header, main, section, article, image, caption, and footer.
3. Save the file and open it in a browser.
4. Identify the gallery container and the six gallery items.
5. Check that every image has an alt attribute describing its content.

ACTIVITY 2 — EXPERIMENT WITH FLEXBOX

1. Find .gallery in the CSS.
2. Test justify-content values: flex-start, center, flex-end, space-between, space-around, and space-evenly.
3. Restore center. Change gap from 16px to 30px, compare, then restore 16px.
4. Change flex-wrap: wrap to nowrap. Narrow the browser and observe what happens. Restore wrap.
5. Record: Why is flex-wrap: wrap useful for an image gallery?

ACTIVITY 3 — CONTROL CARD SIZE

1. Find flex: 1 1 220px in .gallery-item.
2. Change 220px to 280px. Observe how many cards fit in each row.
3. Change max-width: 320px to max-width: 260px. Compare card widths.
4. Restore the original values.
5. Record: What is the effect of changing the flex-basis value?

ACTIVITY 4 — STYLE THE IMAGES

1. Find .gallery-item img.
2. Change height: 200px to height: 250px.
3. Test object-fit: contain, fill, and cover. Compare the results.
4. Restore height: 200px and object-fit: cover.
5. Record: Which object-fit value fills the image area while maintaining proportions?

ACTIVITY 5 — TEST RESPONSIVE DESIGN

1. Find the @media (max-width: 600px) rule.
2. Change 600px to 700px and resize the browser around that width.
3. Temporarily remove flex-direction: column from .gallery inside the media query. Observe the difference.
4. Restore flex-direction: column and align-items: center.
5. Test desktop, tablet-like, and narrow mobile widths.
6. Record: What is a breakpoint, and why should a gallery be tested at different widths?

MINI-PROJECT — MY ICT PHOTO GALLERY

Create a gallery with at least twenty images related to yourself, school, ICT, programming, technology, or student projects.

☐ Use a section with class gallery and article elements with class gallery-item.

☐ Use Flexbox with flex-wrap: wrap and a consistent gap.

☐ Give every image meaningful alt text and a caption.

☐ Use object-fit: cover and a consistent image height.

☐ Add a media query for small screens.

☐ Use a maroon, white, and light-gray color theme.

☐ Add a page title and footer.

☐ Check that cards do not overlap and there is no unwanted horizontal scrolling.

CHECKLIST AND REVIEW

☐ My index.html opens correctly in a browser.

☐ My gallery contains at least eight images.

☐ Cards wrap into additional rows when needed.

☐ Images have consistent display dimensions and meaningful alt text.

☐ The gallery adapts to narrow screens.

1. What is the role of display: flex in the gallery?
   display: flex;
2. What does flex-wrap: wrap do?
   breaks the items to the next line when it exeeds the size of the main container
3. What is the purpose of gap?
   adds gaps between items
4. What is the difference between width: 100% and object-fit: cover?
   width: 100% is used when you want to stretch the image to fill the container, while object-fit: cover is used when you want to maintain the image's aspect ratio
5. Why should images have meaningful alt text?
   to make it easier for screen readers to describe what the image is
6. What does a media query do in this project?
   detects the size of the viewport so that it can adjust the flex layouts

SUBMISSION AND REFERENCE

Suggested filename: Grade11_Name_Responsive_Image_Gallery.html

W3Schools CSS Flexbox: https://www.w3schools.com/css/css3_flexbox.asp

W3Schools Flexbox Container: https://www.w3schools.com/css/css3_flexbox_container.asp

W3Schools Responsive Flexbox: https://www.w3schools.com/css/css3_flexbox_responsive.asp